MATB24H3
Exercise Questions

Student Name: Your Name **Student Number: Your Student Number**

2.1.1. Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ an associative binary operation?

Solution:

2.1.2. Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? what is the identity element? How about $(\mathcal{P}(X), \cup)$ for any arbitrary set X?

Solution:

2.2.1. Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}), +, \cdot)$ forms a field. What happens if $d = m^2$ for some m?

Solution:

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

Solution:

2.2.3. Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and the additive inverse element $-a$ of any element $a \in F$ are unique.

Solution:

2.2.4. Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

Solution:

2.2.5. Prove that the triple $(\mathbb{Z}_p, + \mod p, \cdot \mod p)$ is a field $\iff$ p is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

Solution:

2.2.6. Prove that the 2-tuple $(S, \circ)$, where S is the set of rotational symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

Solution:

2.2.7. Characterise all idempotent elements of $(\mathcal{M}_{2\times 2}(\mathbb{R}), +, \cdot)$.

Solution:

2.2.8. Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

Solution:

2.2.9. Let $(\mathcal{R}, +, \cdot)$ be a ring and suppose that $\alpha \in \mathcal{R}$ has a multiplicative inverse. Moreover, suppose that $\alpha \in C(\mathcal{R})$. Is $\alpha^{-1} \in C(\mathcal{R})$?

Solution:

2.2.10. Does the set of functions with infinitely many roots form a subring of $\mathcal{F}(\mathbb{R})$? Justify your answer.

Solution:

2.2.11. There is no function $g \in \operatorname{supp}(\mathcal{F}(\mathbb{R}))$ that acts as multiplicative identity for all elements in $\operatorname{supp}(\mathcal{F}(\mathbb{R}))$. Can you prove this?

Solution:

2.2.12. Determine the zero and the identity elements of the polynomial ring $(\mathcal{R}[x], +, \cdot)$ where $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2\times 2}(\mathbb{R}))$.

Solution:

2.2.13. True/False. Justify:

1. **Let $\mathbb{F}$ be any field. Suppose that α and β are two elements of an extension field $\mathbb{K}$ of $\mathbb{F}$. Then $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are always distinct extension fields of $\mathbb{F}$.**
2. **Given any set X, there is no operation $\cdot$ that can be defined on $[X]^{\leq|X|}$ that makes it a group.**
3. **Given a monoid $(\mathcal{S}, *)$ there exists a binary operation $\cdot$ on $\mathcal{S}$ such that $(\mathcal{S}, \cdot)$ no longer admits an identity element. If so, what assumption did you need to make?**
4. **$(\mathbb{R}, \cdot)$ forms a group.**
5. **$(\mathbb{R}, +)$ forms a group with the identity element being 1.**
6. **Suppose $(\mathbb{F}, +, \cdot)$ is the field. Then $(\mathbb{F}, +)$ and $(\mathbb{F}, \cdot)$ both form a group.**

Solution:

3.1.1. Had we defined vector addition by $\hat{\boldsymbol{A}} \boxplus \hat{\boldsymbol{B}} = [\boldsymbol{a_{ij}} \cdot \boldsymbol{b_{ij}}]$, i.e., pointwise multiplication of the entries in Example 3.1.3, would $(\boldsymbol{\mathcal{M}_{m\times n}}, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ form a vector space over $\mathbb{F}$?

Solution:

3.1.2. Prove properties 2, 4 and 5 of Theorem 3.1.2, namely, for any vector space $\boldsymbol{V} = (\boldsymbol{V}, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ over field $\mathbb{F}$:

2. **The additive inverse of any $\boldsymbol{v} \in \boldsymbol{V}$ is unique.**
4. **Any scalar of the zero vector yields the zero vector, i.e. $\forall\, \alpha \in \mathbb{F}$, $\boldsymbol{a} \odot \vec{\mathbf{0}} = \vec{\mathbf{0}}$.**
5. **For all $\boldsymbol{v} \in \boldsymbol{V}$ and $\boldsymbol{c} \in \mathbb{F}$, $(-\boldsymbol{c}) \odot \boldsymbol{v} = \boldsymbol{c} \odot (-\boldsymbol{v}) = -\mathbf{1} \odot (\boldsymbol{c} \odot \boldsymbol{v})$.**

Solution:

3.2.1. Prove Theorem 3.2.2, namely if $S \subseteq V$ then $\operatorname{span}(S) \subseteq V$ is a subspace of V.

Solution:

3.2.2. Prove Theorem 3.2.3, namely: Let J be any index set (It is a set whose elements are used to enumerate the elements of another set.) and let $\{W_\alpha \mid \alpha \in J\}$ be a collection of subspaces of V. Then $\bigcap_{\alpha \in J} W_\alpha$ is a subspace of V.

Solution:

3.2.3. Prove Theorem 3.2.4, namely: Let $W_1 \subseteq W_2 \subseteq W_3 \subseteq \ldots$ be an increasing chain of subspaces of V. Then $\bigcup_{n \in \mathbb{N}} W_n$ is a subspace.

Solution:

3.3.1. Show that the set $\{e^x, e^{2x}, e^{3x}\}$ is linearly independent in the vector space of $\mathcal{F}(\mathbb{R})$ over $\mathbb{R}$.

Solution:

3.4.1. Prove Proposition 3.4.1, namely, the following:
Let $T\colon V \to W$ be a linear transformation from V to W. Then:

1. **For all x, $y \in V$, $T(x-y) = T(x) - T(y)$**

2. **T maps the zero vector $\vec{0}_V$ of V to the zero vector $\vec{0}_W$ of W.**

3. **If $T'\colon W \to Z$ is also a linear transformation, their composition $T' \circ T$ is also a linear transformation from V to Z where $T' \circ T = T'(T(v))$. So for every vector $\vec{v} \in V$, we have $(T' \circ T)(\vec{v}) := T'(T(\vec{v}))$.**

4. **If V_1 is a subspace of V and W_1 is a subspace of W, then $T[V_1]$ is a subspace of W and $T^{-1}[W_1]$ is a subspace of V.**

Solution:

3.4.2. Prove

$$\boldsymbol{T : V \to W} \textbf{ is injective} \iff \textbf{ker}\,\boldsymbol{T} = \{\vec{\mathbf{0}}\}$$

Solution:

3.4.3 $(\mathcal{L}(V,W), \boxplus, \odot, (\mathbb{F}, +, \cdot))$ forms a vector space over $\mathbb{F}$, where $\boxplus$ and $\odot$ are the pointwise addition of linear maps and scalar multiplication with the linear map in W, respectively.

That is given $T, S \in \mathcal{L}(V,W)$ and $\alpha \in \mathbb{F}$

$$T \boxplus S := T \underbrace{+}_{W} S$$

$$\alpha \odot T := \alpha \underbrace{\cdot}_{W} T$$

So, given any $v \in V$ the image of v under the linear maps $T \boxplus S$ and $\alpha \odot T$ would be $T(v) +_W S(v)$ and $\alpha \cdot_W T(v)$ respectively.

That is, $\mathcal{L}(V,W)$ is closed under both $\boxplus$ and $\odot$. That is to show that $T \boxplus S$ and $\alpha \odot T$ are indeed linear transformations from V to W and all the axioms of a vector space are satisfied.

Solution:

3.4.4. Prove:
Let V and W be two vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and $\dim(W) = m$. Then, $\dim(\mathcal{L}(V, W)) = mn$

Hint: Note that a linear tranformation T is determined by its action on its basis elements of its domain. So, if $\mathcal{B}$ is a basis for V over $\mathbb{F}$, and $T \in \mathcal{L}(V, W)$, then as long as we know what $T(b)$ is for every $b \in \mathcal{B}$, (the basis could be finite or infinite) we could then determine the image of an arbitrary vector $v \in V$. Since $v = \sum_{i=1}^{n} \alpha_i b_i$ for some $n \in w$, $b_i \in \mathcal{B}$, and $\alpha_i \in \mathbb{F}$, then by the properties of linear operators we have:

$$T(v) = \sum_{i=1}^{n} \alpha_i T(b_i)$$

Solution:

3.4.5. Prove that a linear map $T\colon V \to W$ is invertible $\iff$ T is injective and surjective.

Solution:

3.4.6. Let V and W be two vector spaces over a field $\mathbb{F}$ with dimensions n and $m > 3$ respectively. Construct three different linearly independent maps from V to W.

Solution:

3.4.7. Prove whether or not the triple $(\mathcal{L}(\mathbb{R}, \mathbb{R}), +, \cdot)$ forms a subring of $\mathcal{F}(\mathbb{R})$ under point-wise addition and multiplication.

Solution:

3.6.1. Let V and W be two vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and $\dim(W) = m$. Then $\dim(\mathcal{L}(V, W)) = mn$

Solution:

3.6.2. Suppose A and B are two similar matrices of size n. Prove whether or not they represent, or can be viewed as matrix representations of the same linear transformation relative to suitable ordered bases.

Solution:

3.6.3. True/False. Justify

1. **Let $\mathcal{B}$ and $\mathcal{B}'$ be two ordered bases for the vector space $\mathbb{R}^n$ over $\mathbb{R}$. Then $\det(\mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}) = 1$**
2. **For all choices of ordered bases $\mathcal{B}$ and $\mathcal{B}'$ for $\mathbb{R}^n$, $\det(\mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}) \neq 0$.**
3. **Let $T \in \mathcal{L}(V, W)$ be two vector spaces of dimension n over a field $\mathbb{F}$. If T is surjective, then it must be an isomorphism.**
4. **$\mathbb{P}_{10}[x] \cong \mathbb{R}^{11}$.**
5. **$T \in \mathcal{L}(V, W)$ is an isomorphism $\iff \dim(V) = \dim(W)$.**
6. **There are finite dimensional vector spaces V and W such that $\dim(\mathcal{L}(V, W)) = 32$.**

Solution:

4.2.1 For a basic vector space V over $\mathbb{C}$, prove the following property of the inner product:

$$\forall u, v, w \in V, \langle v + u, w \rangle = \langle v, w \rangle + \langle u, w \rangle$$

Solution:

4.2.2 Verify the following function is an inner product on $\mathcal{C}_0([0,1])$

$$\langle \cdot, \cdot \rangle : \mathcal{C}_0([0,1]) \times \mathcal{C}_0([0,1]) \to \mathbb{R}$$

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)dx$$

Solution:

4.2.3 Verify the following function is an inner product on $\mathbb{C}^n$

$$\vec{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_n), \quad \vec{\beta} = (\beta_1, \beta_2, \ldots, \beta_n) \in \mathbb{C}^n$$

$$\langle \vec{\alpha}, \vec{\beta} \rangle = \sum_{i=1}^{n} \overline{\alpha_i} \beta_i$$

Solution:

4.2.4 Could we have defined the inner product on $\mathbb{C}^n$ by $\langle \vec{\alpha}, \vec{\beta} \rangle = \sum_{i=1}^{n} \alpha_i \overline{\beta_i}$. Would it be an inner product over $\mathbb{C}^n$?

Solution:

4.2.5. Let $(V, \langle \cdot, \cdot, \rangle)$ be an inner product space. Then V is a normed vector space with norm $\|v\| = \sqrt{\langle v, v \rangle}$. Prove the positive-definiteness of the norm.

Solution:

4.2.6. Could we ever obtain equality in the Cauchy-Schwarz inequality? i.e,

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \sqrt{\langle \boldsymbol{u}, \boldsymbol{u} \rangle} \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$$

Solution:

4.2.7. Prove the generalised version of the Pythagorean Theorem. i.e., Let V be an inner product space. Let u, v be any two orthogonal vectors in V. Then:

$$\|u_v\|^2 = \|u\|^2 + \|v\|^2$$

Solution:

4.2.8. Define the discrete metric d on a vector space V as follows:

$$\forall \boldsymbol{u}, \boldsymbol{v} \in \boldsymbol{V}, \quad \boldsymbol{d}((\boldsymbol{u}, \boldsymbol{v})) = \begin{cases} \mathbf{0} & \textbf{if } \boldsymbol{u} = \boldsymbol{v}, \\ \mathbf{1} & \textbf{otherwise.} \end{cases}$$

Verify that d is indeed a metric on V.

Solution:

4.2.9. In Example 4.2.9, we define a metric on $\mathcal{C}_0([0,1],\mathbb{R})$ as follows:

$$d(f(x), g(x)) \longmapsto \int_0^1 |f(x) - g(x)|\psi(x)dx$$

(where $\psi : [0,1] \to (0,\infty)$ is any fixed continuous function and $\psi \neq 1$)

Prove that function d is indeed a metric on $\mathcal{C}_0([0,1],\mathbb{R})$. Make sure you understand it well. What happens if ψ is not continuous?

Solution:

4.2.10. Let $(V, \|\cdot\|)$ be a normed vector space. Also, let $\|\cdot\|$ follow the paralellogram law:

$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

Define the inner product on V as:

$$(u, v) \longmapsto \langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u-iv\|^2 - i\|u+iv\|^2\right)$$

Prove that $\langle\cdot,\cdot\rangle$ is indeed an inner product on V.

Solution: